

SynthCCD & NENA Performance Alignment Guide

Systematic Synchronization of Synthetic Data Models with Public Safety Metrics Standards

1. Operational Alignment Framework

Modern Public Safety Answering Points (PSAPs) rely heavily on Computer-Aided Dispatch (CAD) and Telephony/CPE logging databases to audit compliance with critical performance benchmarks. The integration of **SynthCCD** (a synthetic dispatch and hourly phone-metric generator) with the NENA Performance Audit Template establishes a robust, closed-loop testing environment. By simulating highly realistic dispatch intervals, personnel scheduling, and queuing metrics, agencies can evaluate automated reporting tools, train administrative staff, and stress-test reporting dashboards without compromising real citizen data or exposing Protected Health Information (PHI).

This guide outlines the concrete technical recommendations and data schema enhancements required to maximize the utility of SynthCCD in 9-1-1 center performance audits. Aligning these two platforms enables public safety authorities to develop rigorous contingency models, conduct accurate staffing audits, and deploy unified, cross-vendor reporting templates that match NENA-STA-019.2-2022 and NENA-STA-020.2-2026 standards.

2. Core Data Schema Gaps & Integration Recommendations

While the current SynthCCD version models standard dispatch and hourly counters effectively, several crucial data gaps must be addressed to enable seamless compatibility with an advanced NENA performance audit.

NENA Performance Dimension	Data Schema Gaps in SynthCCD	Recommended Schema Extensions
High Impact Events (HIEs) Exclusion (NENA-STA-020.2, Sec. 7)	No standard column to isolate crisis spikes (disasters, active threats) from routine operations.	Add <code>high_impact_event_flag</code> (Boolean) and <code>hie_type</code> (Categorical) to exclude HIEs dynamically from monthly audits.
NG9-1-1 SIP Signaling Delays (NENA-STA-019.2, Sec. 3.3)	Lacks transit telemetry inside the Emergency Services IP Network (ESInet).	Introduce <code>esinet_transit_seconds</code> , <code>sip_ssrdd_seconds</code> (SSRD), and <code>sip_sdd_seconds</code> (SDD) to audit network routing latency.
Special Call Typology & TTY (NENA-STA-020.2, Sec. 5.4)	Abandoned counts are only tracked hourly; individual CAD records lack call-handling classifications.	Append <code>call_handling_type</code> (Standard, Abandoned, Silent, Non-Responsive) and <code>tty_interrogated_flag</code> (Boolean).
Agent State Tracking (NENA-STA-019.2, Sec. 3.4)	Individual state transitions (LoggedOut, Break, Waiting, Active) are absent.	Expose an auxiliary <code>agent_state_history</code> log table linked to personnel IDs to calculate primary vs. secondary states.
Warm Call Transfers (NENA-STA-020.2, Sec. 5.3)	Inter-PSAP routing and warm/attended transfer details are unrepresented.	Add <code>transferred_to_agency</code> (String) and <code>warm_transfer_overlap_seconds</code> to verify dispatcher overlap policies.
Media Quality Assurance (NENA-STA-019.2, Sec. 3.3)	No telemetry for voice path quality.	Include <code>media_jitter_ms</code> , <code>media_packet_loss_pct</code> , and <code>mos_score</code> to track performance across network borders.

3. Realism Tuning Workflows for 9-1-1 Administrators

Public safety administrators can programmatically adjust SynthCCD's statistical engines to emulate varied dispatch environments. The generator leverages lognormal distributions that represent the skewed nature of real emergency events (where most actions happen quickly, but a narrow 'tail' of events exhibits extreme delays).

Step-by-Step Calibration Protocol:

- **Extract Empirical Baseline Statistics:** Query at least 6 months of historical CAD data to determine the 50th and 90th percentile times for call pickup, dispatcher interview, vehicle turnout, and transit.
- **Convert to Lognormal Parameters:** Translate empirical averages into the mu (μ) and sigma (σ) parameters required by SynthCCD's configuration schema.
- **Calibrate Telephony Performance:** Edit the phone_metrics section of the realism YAML. Lowering nine_one_one_answer_time_mu to 1.10 concentrates pickup times to 3 seconds, guaranteeing standard compliance.
- **Model Dispatching Resource Gaps:** Adjust shift_config to define unique day/night rosters. Reducing the calltaker/dispatcher staffing pool in underfunded schedules will trigger automatic answer-time bottlenecks.
- **Audit and Re-run:** Save the custom YAML configuration and launch a small-scale run (--rows 1000 --format pandas) to check that the simulated output correlates with the target baseline within $\pm 5\%$ tolerances.

Operational Insight: When modeling rural centers, administrators should adjust the time_profiles.travel_mean parameter up to 600.0 (10 minutes) and turnout_mean to 90.0 to accurately simulate volunteer-staffed fire stations and extended transit delays.

4. Mitigating Time Metrology and Timestamp Rounding Errors

A critical pitfall in public safety database systems is **timestamp truncation**. Many operational systems record timestamps (e.g., time_phone_pickup or time_call_enters_queue) only to the nearest minute. When an analyst attempts to calculate call processing times by subtracting these timestamps, the resulting values are severely distorted by minute-level rounding (resulting in artificial intervals like exactly 60 or 120 seconds).

The Metrology Solution: SynthCCD resolves this programmatically. Although its datetime columns comply with standard database formats and may exhibit minute-level rounding, the generator exports **explicit interval columns** calculated at the sub-second level (such as pickup_delay_seconds, interview_seconds, and turnout_seconds).

Key Directive for Reporting Teams: Always write your automated compliance dashboards and SQL reporting scripts to query these pre-calculated, explicit seconds columns. Using the raw datetime columns for mathematical differences will artificially deflate compliance rates and lead to incorrect performance auditing assessments.

5. Recommended Compliance Stress-Testing Scenarios

To develop a highly resilient NENA-compliant reporting framework, agencies should utilize different SynthCCD configurations to simulate three primary stress-testing baselines:

Scenario Profile	YAML Parameters to Adjust	Expected Performance Audit Outcome
High-Performance Baseline (PASS)	Lower nine_one_one_answer_time_mu to 1.10. Set interview_mean to 20.0 (P1) and 30.0 (P2-5).	9-1-1 answering speed achieves >90% compliance under 15s. Call processing compliance exceeds 85%.

Scenario Profile	YAML Parameters to Adjust	Expected Performance Audit Outcome
Systemic Staffing Deficit (FAIL)	Raise nine_one_one_answer_time_mu to 2.20. Reduce shift_config calltakers from 4 down to 1.	Answering speed collapses below 70% under 15s. Abandoned call rates spike dramatically during busy hours.
Severe Dispatch Friction	Increase dispatch_mean to 90.0. Set turnout_mean to 85.0.	Answering speed remains compliant, but CAD processing audits fail NENA's 60s/90s dispatch standards.

Conclusion & Best Practices: By pairing SynthCCD's flexible parameter scaling with our unified NENA Performance Audit Workbook, public safety agencies can establish a world-class Quality Assurance loop. This allows reporting teams to systematically verify that their dashboards, personnel benchmarks, and network configurations are fully prepared to meet modern NG9-1-1 requirements.